Healthcare Duplicate MRN Continuous Improvement Analysis

Using Python, SQL, and Power BI to Identify Opportunities for Reducing Duplicate MRNs

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Healthcare Analytics Portfolio

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What problem are we trying to solve?

Business Problem

Duplicate medical record numbers (MRNs) create operational inefficiencies, increase duplicate investigation workload, and can contribute to patient safety risks if patient information is fragmented across multiple records.

Project Goal

Identify the highest-impact process improvement opportunities for reducing duplicate MRNs through analysis of duplicate investigation data.

How was the solution developed?

Workflow

- Business Problem
- Domain Knowledge
- Python Simulation
- SQL Analysis
- Power BI Dashboard
- Recommendations

Technology Stack

- Python (data simulation)
- SQL (business analysis)
- Power BI (visualization)

Why synthetic data?

Synthetic data was modeled after realistic healthcare registration workflows to demonstrate an end-to-end analytics solution while protecting patient privacy.

How was the solution built?

Python Data Generation

- 1,000 synthetic duplicate investigations
- Weighted probability simulation
- Source-specific business rules

Healthcare Workflow Rules

- Mutually exclusive contributing factors
- Realistic healthcare registration workflows
- Categorized contributing factors into five primary categories

SQL Analysis

- Business question-driven SQL queries
- Aggregated trends using GROUP BY, COUNT, and ORDER BY
- Summarized trends by source, department, contributing factor, and month

Power BI

- Two-page executive dashboard
- Interactive Opportunity Explorer
- Continuous improvement-focused reporting

What did the analysis reveal?

Total Duplicate MRNs

1,000

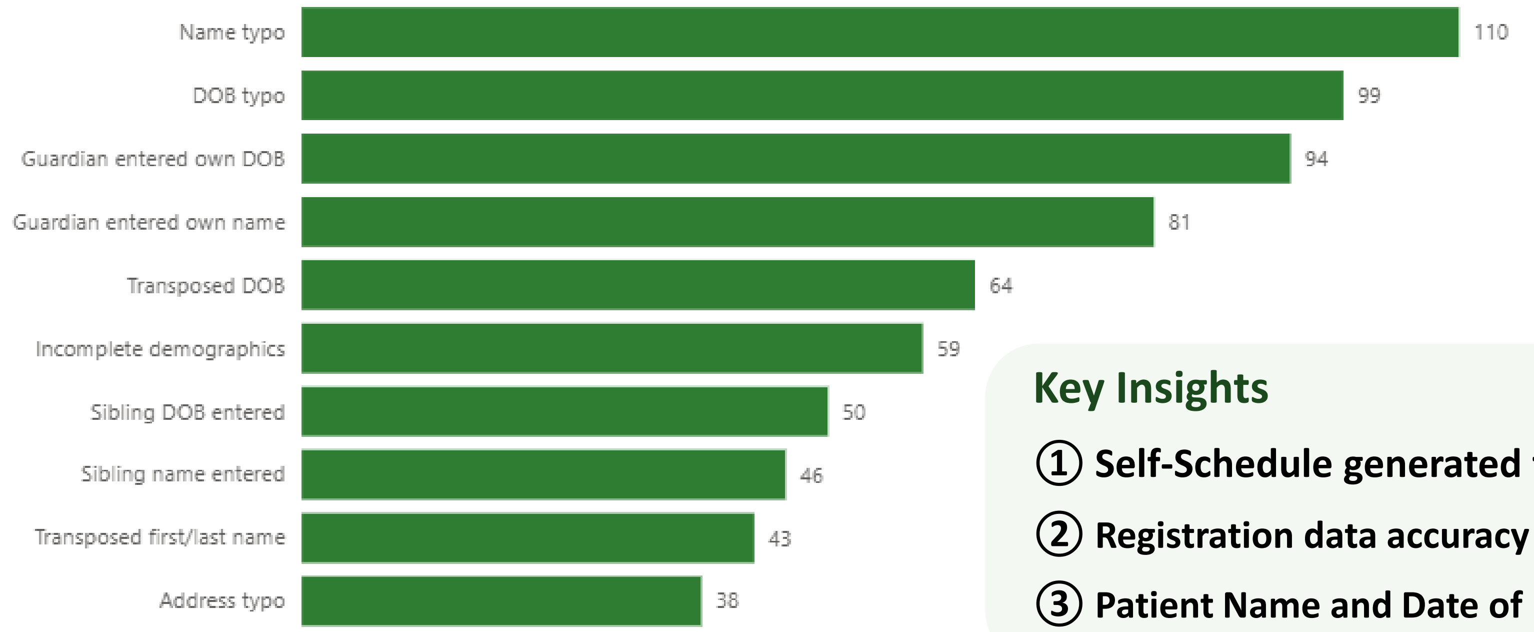
Primary Duplicate Source

Urgent Care (34.5%)

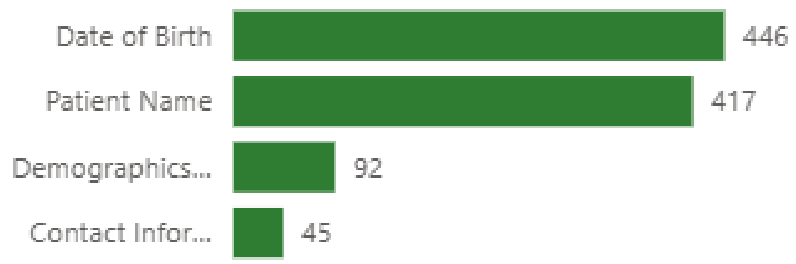
Top Contributing Factor

Address typo

Top Improvement Opportunities



Contributing Factor Categories



Key Insights

① **Self-Schedule generated the highest duplicate volume (34.5%).**

② **Registration data accuracy represents the largest improvement opportunity.**

③ **Patient Name and Date of Birth accounted for most duplicate investigations.**

How can stakeholders explore improvement opportunities?

Opportunity Explorer

Enables department leaders to investigate duplicate MRN patterns through interactive filtering and drill-down analysis.

Capabilities

✓ Compare duplicate trends by source

✓ Identify high-volume departments

✓ Explore service-line trends

✓ Monitor trends over time

✓ Review investigations details

Duplicate MRN Opportunity Explorer

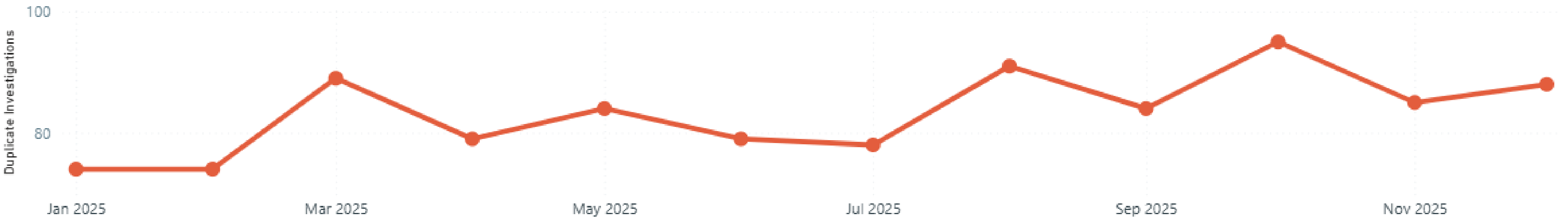
Explore improvement opportunities through interactive filtering and trend analysis

Source: All

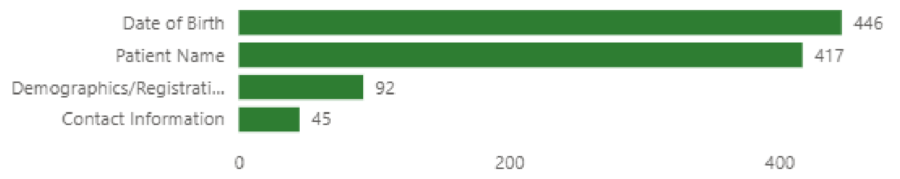
Department: All

Service Line: All

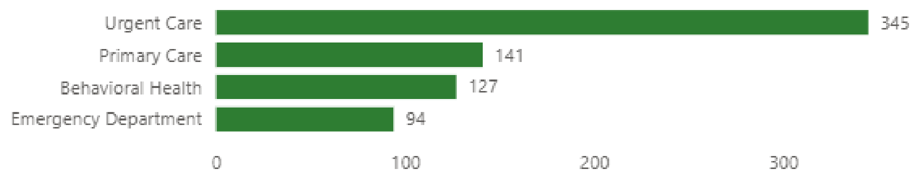
Duplicate Investigations Over Time



Primary Categories



Department Breakdown



Duplicate Investigation Details

Duplicate ID	Event Date	Source	Department	Service Line	Primary Category	Contributing Factors	Status	Resolution Type
1	Wednesday, October 01, 2025	Internal Registration	Primary Care	Ambulatory	Date of Birth	DOB typo	Resolved	Merge
2	Tuesday, April 01, 2025	Self-Schedule	Urgent Care	Ambulatory	Patient Name	Guardian entered own name	Resolved	Merge
3	Saturday, March 01, 2025	Internal Registration	Lab	Ancillary	Patient Name	Transposed first/last name	Resolved	KND
4	Saturday, February 01, 2025	Internal Registration	Emergency Department	Emergency	Demographics/Registration	Incomplete demographics	Resolved	Merge
5	Tuesday, July 01, 2025	Self-Schedule	Urgent Care	Ambulatory	Patient Name	Guardian entered own name	Resolved	Merge

Recommended Process Improvements

Recommendations aligned to the highest-impact findings

Improve Self-Schedule Registration

Collaborate with the Self-Schedule team to simplify parent registration workflows and clarify field instructions.

Addresses the highest-volume duplicate source.

Reinforce Reliable Search Method (RSM)

Reinforce the RSM for internal registration staff and external providers through refresher education and targeted reminders.

Reduces duplicate creation before new MRNs are generated.

Reinforce Registration Safety Tools

Reinforce use of safety tools such as **STAR (Stop, Think, Act, Review)** and **Repeat-Back** during registration to verify patient information before completing registration.

Improves registration data accuracy.

Targeted Duplicate MRN Training

Require frequent duplicate creators to complete the "Duplicate MRNs & Overlays" learning course.

Provides targeted coaching based on performance.

Monitor Department Performance

Provide department leaders with a real-time dashboard and conduct quarterly quality improvement rounding to monitor duplicate trends.

Supports continuous improvement and accountability.

Thank you / Questions

Thank you for your time and feedback.

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